DEEP REINFORCEMENT LEARNING
Incremental Example

DRL Team

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The instructors is gratefully acknowledging the authors who made their course materials freely available online.

This deck is designed by BITS WILP faculty team of course Deep Reinforcement Learning

Table of Contents

1. MDP
2. DP
3. MC
4. TD (to be updated)

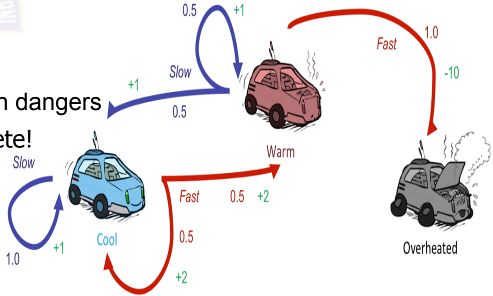
Problem Definition

Race Car Laps

A race car completes laps while managing engine temperature and speed decisions. Here each step is a segment of a lap in the track where the car learns to trades off lap time against overheating risk. The more the lap the car completes it performance betters. Car shuts down only when its engine get overheated (optional terminal state).

Key Characteristics:

- The agent must act under **uncertainty** —
  percepts are local, and the agent must reason about hidden dangers
- Assume there is no fuel constraints or No.of.Laps to complete!
- At every time step the next potential engine temperature depends on the current engine state not the past history



MARKOV DECISION PROCESS FORMALIZATION (Simplified)

1. States (S)
   - S = (Cool(**C**), Warm(**W**), Overheated(**O**)) #Engine Temperature
   - Finite State MDP
2. Actions (A)
   - A = {Go Slow(**S**), Go Fast **(F) }**
3. Rewards (R)
   - $$R(s,a,s') = \begin{cases} +1 & \text{if } s' \text{ is reached by going } \mathit{Slow} \\ +2 & \text{if } s' \text{ is reached by going } \mathit{Fast} \\ -10 & \text{if } s' \text{ is terminal} \end{cases}$$
4. State Transition Probabilities (P)
   - $$P(s'|s,a) = \begin{cases} 0.5 & \text{if action lead to different non} - \text{terminal state} \\ 1 & \text{otherwise} \end{cases}$$

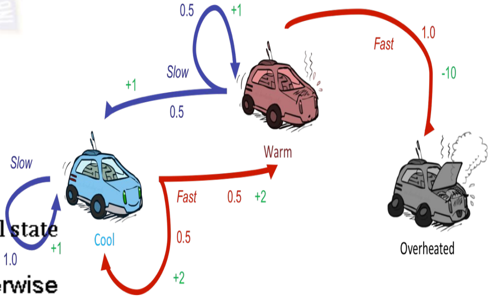


Table of Contents

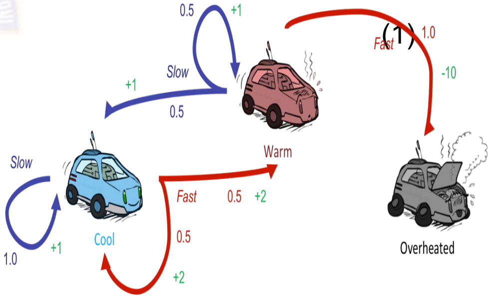
1. MDP
2. **DP**
3. MC
4. TD (to be updated)

Dynamic Programming

Scenario

In a modified Race car, there is a training track where agent is given a known map and risk factors(dynamics) in advance. Agent knows exactly where it is and its action how actions result in engine state change (KNOWN MODEL).

With full knowledge, the agent can compute optimal policies by trying through all states.



Dynamic Programming

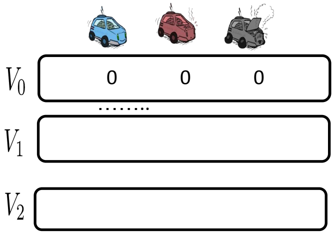
1. Given
   - P(s` | s, a) , R(s, a, s`) , P(s`, r | s, a)
2. To find
   - Policy π
3. Solution
   - Policy Iteration (PI): Prefer this in a relatively smaller state space where intermediate termination provides explicit interpretable policy
   - Value Iteration (VI): Prefer this in a relatively larger state space where fast convergence to optimal policy is obtained at the end
4. Idea
   - PI : One Step Look ahead □ Evaluate □ Improve
   - VI : One Step Look ahead □ Control

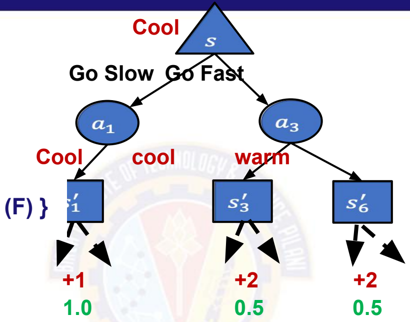
VI can be thought of truncated PI where evaluation is done fully before improvement.

Dynamic Programming Solution

Value Iteration Algorithm

Dynamic Programming – Value Iteration





Value iteration

Initialize array V arbitrarily (e.g., $V(s) = 0$ for all $s \in \mathcal{S}^+$)

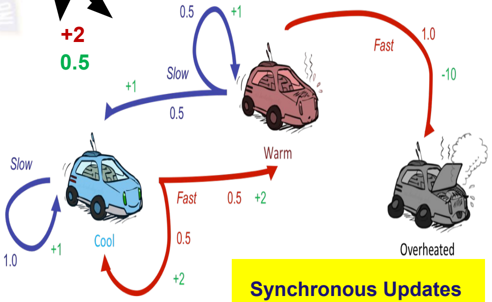
$\gamma = 0.9$

$\theta = 0.01$

Repeat
- $\Delta \leftarrow 0$
- For each $s \in \mathcal{S}$:
  - $v \leftarrow V(s)$
  - $V(s) \leftarrow \max_a \sum_{s',r} p(s',r|s,a)\big[r + \gamma V(s')\big]$
  - $\Delta \leftarrow \max(\Delta, |v - V(s)|)$

until $\Delta < \theta$ (a small positive number)

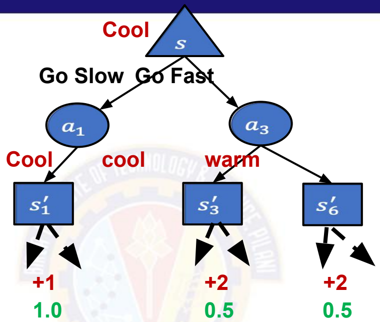
Output a deterministic policy, $\pi \approx \pi_*$, such that
$\pi(s) = \operatorname{argmax}_a \sum_{s',r} p(s',r|s,a)\big[r + \gamma V(s')\big]$



Synchronous Updates

Dynamic Programming – Value Iteration

V_0	0	0	0
V_1			
V_2			



Value iteration

Initialize array V arbitrarily (e.g., $V(s) = 0$ for all $s \in \mathcal{S}^+$)

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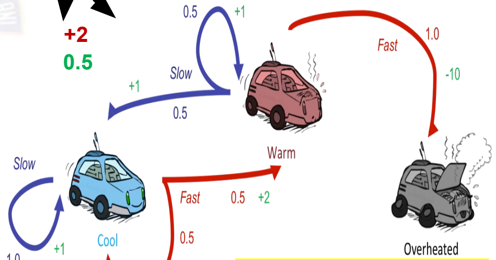


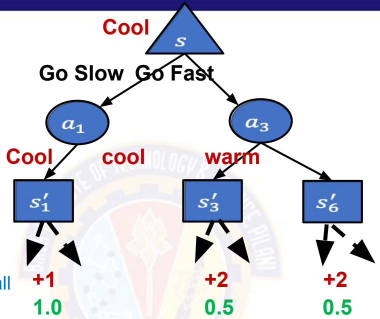
Synchronous Updates

Dynamic Programming – Value Iteration

V_0	0	0	0
V_1	2	1	
V_2			

....

0..........



Value iteration

Initialize array V arbitrarily (e.g., $V(s) = 0$ for all $s \in \mathcal{S}^+$)

$\gamma = 0.9$

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Repeat
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  - $\Delta \leftarrow \max(\Delta, |v - V(s)|)$

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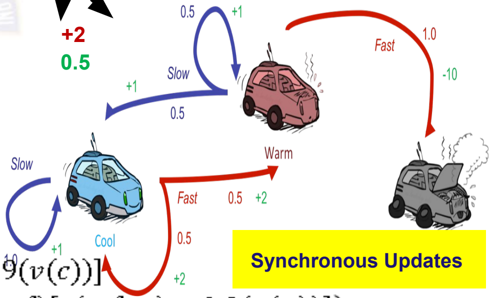
Output a deterministic policy, $\pi \approx \pi_*$, such that
$\pi(s) = \arg\max_a \sum_{s',r} p(s',r|s,a)\left[r + \gamma V(s')\right]$

- Below is done only for state : "Cool". Repeat this for all the states "Warm", "Overheated" per iteration

$$V(cool) = \max \begin{cases} slow: 1 \\ fast: 2 \end{cases}$$

$$V(cool) = \max \begin{cases} 1 * [1 + 0.9(0)] \\ (0.5 * [2 + 0.9(0)]) + (0.5[2 + 0.9(0)]) \end{cases}$$

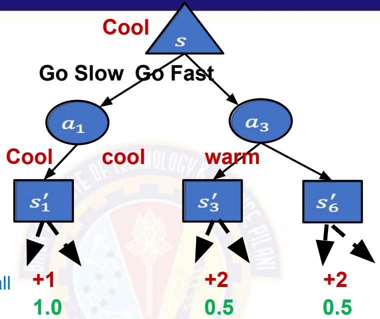
$$V(cool) = \max \begin{cases} P(c\,|c,s)[r(c,s,c) + 0.9(v(c))] \\ (P(c\,|c,f)[r(c,f,c) + 0.9(v(c))]) + (P(w|c,f)[r(c,f,w) + 0.9(v(w))]) \end{cases}$$



Synchronous Updates

Dynamic Programming – Value Iteration

V_0	0	0	0
V_1	2	1	0
V_2	3.35	2.35	0



- Below is done only for state : "Cool". Repeat this for all the states "Warm", "Overheated" per iteration

$$V(cool) = \max\begin{cases} slow: 2.8 \\ fast: 3.35 \end{cases}$$

$$V(cool) = \max\begin{cases} 1 * [1 + 0.9(2)] \\ (0.5 * [2 + 0.9(2)]) + (0.5[2 + 0.9(1)]) \end{cases}$$

$$V(cool) = \max\begin{cases} P(c\,|c,s)[r(c,s,c) + 0.9(v(c))] \\ (P(c\,|c,f)[r(c,f,c) + 0.9(v(c))]) + (P(w|c,f)[r(c,f,w) + 0.9(v(w))]) \end{cases}$$

Value iteration

Initialize array V arbitrarily (e.g., $V(s) = 0$ for all $s \in \mathcal{S}^+$)

$\gamma = 0.9$

$\theta = 0.01$

Repeat
- $\Delta \leftarrow 0$
- For each $s \in \mathcal{S}$:
  - $v \leftarrow V(s)$
  - $V(s) \leftarrow \max_a \sum_{s',r} p(s',r|s,a)\big[r + \gamma V(s')\big]$
  - $\Delta \leftarrow \max(\Delta, |v - V(s)|)$

until $\Delta < \theta$ (a small positive number)

Output a deterministic policy, $\pi \approx \pi_*$, such that
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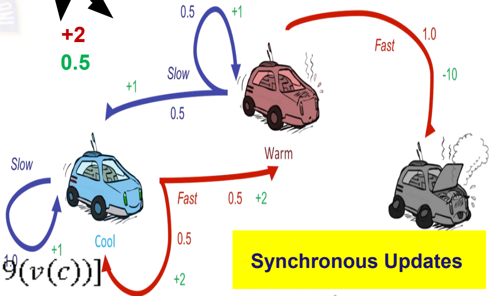
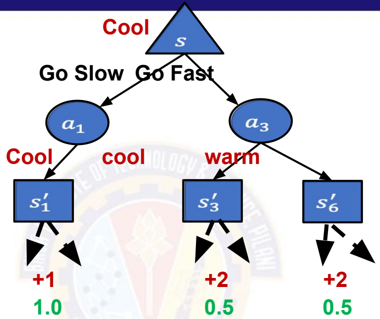


Synchronous Updates

Dynamic Programming – Value Iteration

V_0	0	0	0
V_1	2	1	0
V_2	3.35	2.35	0
V_{49}	~15.41	~14.41	0



Value iteration

Initialize array V arbitrarily (e.g., $V(s) = 0$ for all $s \in \mathcal{S}^+$)

$\gamma = 0.9$

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Repeat
- $\Delta \leftarrow 0$
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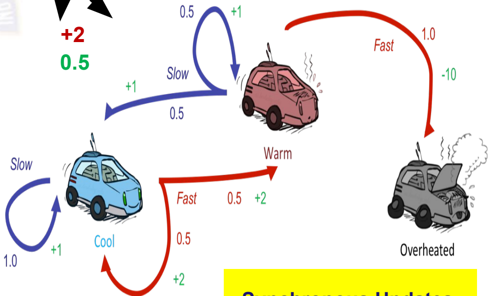
until $\Delta < \theta$ (a small positive number)

Output a deterministic policy, $\pi \approx \pi_*$, such that

$\pi(s) = \operatorname{argmax}_a \sum_{s',r} p(s',r|s,a)\left[r + \gamma V(s')\right]$

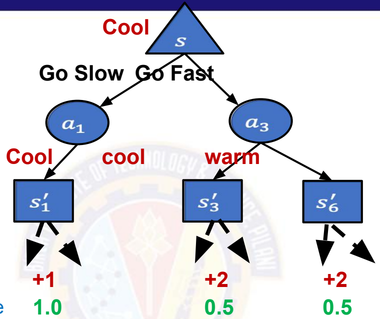


- Δ = ~0.009 less than threshold ϴ = 0.01
- Algorithm converges slowly. Without ϴ converges ~208th
- **Can faster convergence possible ? Yes via Asynchronous Value Updates. Idea is to use the recent values V(S`) even within the same iteration ! One iteration is shown in next slide**

Synchronous Updates

Dynamic Programming – Value Iteration

V_0	0	0	0
....			
V_1	2		
V_2			



Value iteration

Initialize array V arbitrarily (e.g., $V(s) = 0$ for all $s \in \mathcal{S}^+$)

$\gamma = 0.9$

$\theta = 0.01$

Repeat
- $\Delta \leftarrow 0$
- For each $s \in \mathcal{S}$:
  - $v \leftarrow V(s)$
  - $V(s) \leftarrow \max_a \sum_{s',r} p(s',r|s,a)\left[r + \gamma V(s')\right]$
  - $\Delta \leftarrow \max(\Delta, |v - V(s)|)$

until $\Delta < \theta$ (a small positive number)

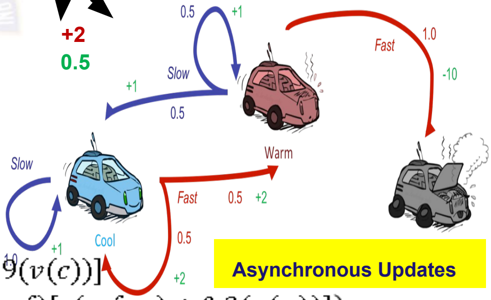
Output a deterministic policy, $\pi \approx \pi_*$, such that
$\pi(s) = \arg\max_a \sum_{s',r} p(s',r|s,a)\left[r + \gamma V(s')\right]$

- Below is done only for state : "Cool". In this same iteration while updating the V(Warm) use recent value of V(Cool) = 2 not 0 .

$$V(cool) = \max \begin{cases} slow: 1 \\ fast: 2 \end{cases}$$

$$V(cool) = \max \begin{cases} 1 * [1 + 0.9(0)] \\ (0.5 * [2 + 0.9(0)]) + (0.5[2 + 0.9(0)]) \end{cases}$$

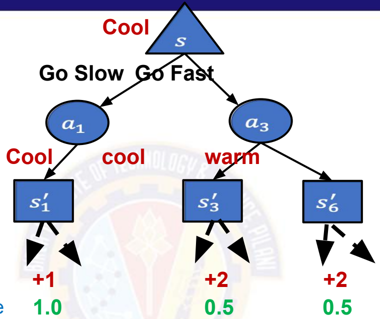
$$V(cool) = \max \begin{cases} P(c|c,s)[r(c,s,c) + 0.9(v(c))] \\ (P(c|c,f)[r(c,f,c) + 0.9(v(c))]) + (P(w|c,f)[r(c,f,w) + 0.9(v(w))]) \end{cases}$$



Asynchronous Updates

Dynamic Programming – Value Iteration

V_0	0	0	0
	....		
V_1	**2**	**1.9**	**0**
V_2			



Value iteration

Initialize array V arbitrarily (e.g., $V(s) = 0$ for all $s \in \mathcal{S}^+$)

$\gamma = 0.9$

$\theta = 0.01$

Repeat

$\quad \Delta \leftarrow 0$

$\quad$ For each $s \in \mathcal{S}$:

$\quad\quad v \leftarrow V(s)$

$\quad\quad V(s) \leftarrow \max_a \sum_{s',r} p(s',r|s,a)\big[r + \gamma V(s')\big]$

$\quad\quad \Delta \leftarrow \max(\Delta, |v - V(s)|)$

until $\Delta < \theta$ (a small positive number)

Output a deterministic policy, $\pi \approx \pi_*$, such that

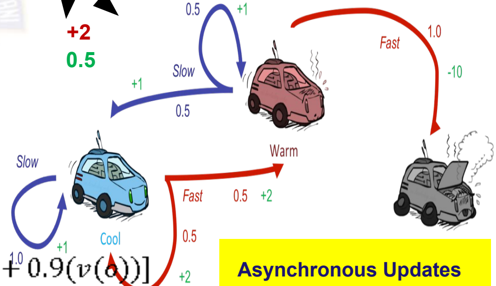
$\quad \pi(s) = \arg\max_a \sum_{s',r} p(s',r|s,a)\big[r + \gamma V(s')\big]$

- Below is done only for state : "Cool". In this same iteration while updating the V(Warm) use recent value of V(Cool) = 2 .

$$V(warm) = \max \begin{cases} fast: -10 \\ slow: 1.9 \end{cases}$$

$$V(warm) = \max \begin{cases} 1 * [-10 + 0.9(0)] \\ (0.5 * [1 + 0.9(2)]) + (0.5[1 + 0.9(0)]) \end{cases}$$

$$V(warm) = \max \begin{cases} P(o\,|w,f)[r(w,f,o) + 0.9(v(o))] \\ (P(c\,|w,s)[r(w,s,c) + 0.9(v(c))]) + (P(w|w,s)[r(w,s,w) + 0.9(v(w))]) \end{cases}$$



Asynchronous Updates

Dynamic Programming – Value Iteration

V_0: 0 0 0

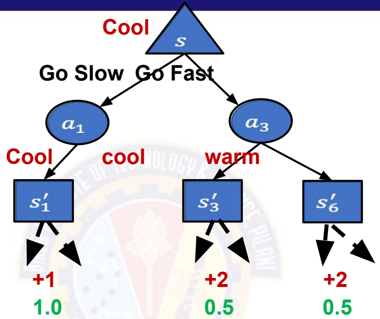
....

V_1: 2 1.9 0..........

V_2: 3.75 3.54 0

....

V_{40}: ~15.44 ~14.44 0



Value iteration

Initialize array V arbitrarily (e.g., $V(s) = 0$ for all $s \in \mathcal{S}^+$)

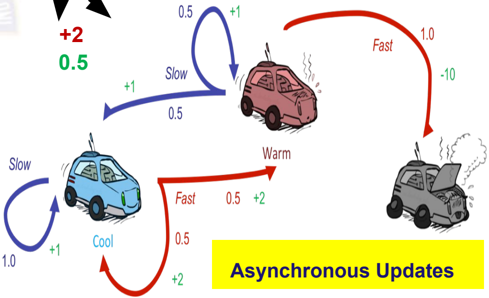
$\gamma = 0.9$

$\theta = 0.01$

Repeat
  $\Delta \leftarrow 0$
  For each $s \in \mathcal{S}$:
    $v \leftarrow V(s)$
    $V(s) \leftarrow \max_a \sum_{s',r} p(s', r|s, a)\big[r + \gamma V(s')\big]$
    $\Delta \leftarrow \max(\Delta, |v - V(s)|)$
until $\Delta < \theta$ (a small positive number)

Output a deterministic policy, $\pi \approx \pi_*$, such that
  $\pi(s) = \arg\max_a \sum_{s',r} p(s', r|s, a)\big[r + \gamma V(s')\big]$

- Δ = ~0.008 less than threshold θ = 0.01
- Algorithm converges fast. Without θ converges at ~157th iteration!



Asynchronous Updates

Dynamic Programming – Value Iteration

V_0: 0 0 0

....

V_1: 2 1.9 0

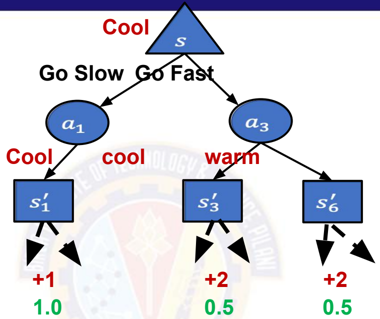
....

V_2: 3.75 3.54 0

....

V_{40}: ~15.44 ~14.44 0

State	Best Action
Cool	Fast
Warm	Slow
Overheated	-



Value iteration

Initialize array V arbitrarily (e.g., $V(s) = 0$ for all $s \in \mathcal{S}^+$)

$\gamma = 0.9$

$\theta = 0.01$

Repeat

  $\Delta \leftarrow 0$

  For each $s \in \mathcal{S}$:

    $v \leftarrow V(s)$

    $V(s) \leftarrow \max_a \sum_{s',r} p(s',r|s,a)\big[r + \gamma V(s')\big]$

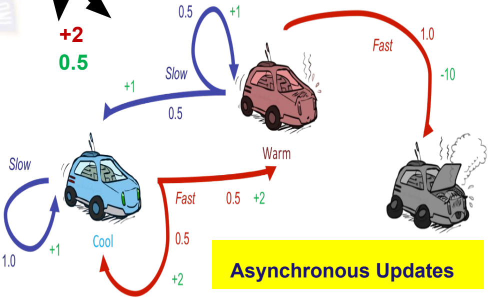
    $\Delta \leftarrow \max(\Delta, |v - V(s)|)$

until $\Delta < \theta$ (a small positive number)

Output a deterministic policy, $\pi \approx \pi_*$, such that

  $\pi(s) = \arg\max_a \sum_{s',r} p(s',r|s,a)\big[r + \gamma V(s')\big]$

- After convergence extract the policy by choosing the action that lead to the converged maximum value here from V40th iteration



Dynamic Programming Solution

In many cases, sometimes policies is observed to converge faster than value!

So why wait for entire value updates?

Policy Iteration Algorithm

$$\pi_0 \xrightarrow{E} v_{\pi_0} \xrightarrow{I} \pi_1 \xrightarrow{E} v_{\pi_1} \xrightarrow{I} \pi_2 \xrightarrow{E} \cdots \xrightarrow{I} \pi_* \xrightarrow{E} v_*,$$

Dynamic Programming

1. Given
   - P(s` | s, a) , R(s, a, s`) , P(s`, r | s, a)
2. To find
   - Policy π
3. Solution
   - Policy Iteration (PI): Prefer this in a relatively smaller state space where intermediate termination provides explicit interpretable policy
   - Value Iteration (VI): Prefer this in a relatively larger state space where fast convergence to optimal policy is obtained at the end
4. Idea
   - PI : One Step Look ahead Evaluate Improve
   - VI : One Step Look ahead Control

 Number of state sweeps (updates) can be controlled if required (similar to mini-batch /stochastic processing). Evaluate & Improve steps are alternatively are done .

VI can be thought of truncated PI where evaluation is done fully before improvement.

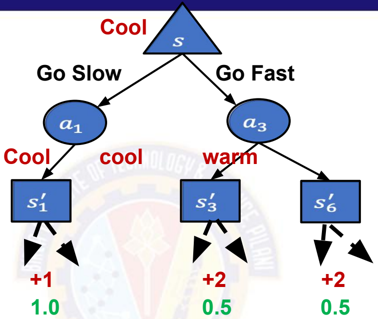
Dynamic Programming – Policy Iteration

V_0 | 0 | 0 | 0

....

V_1

V_2



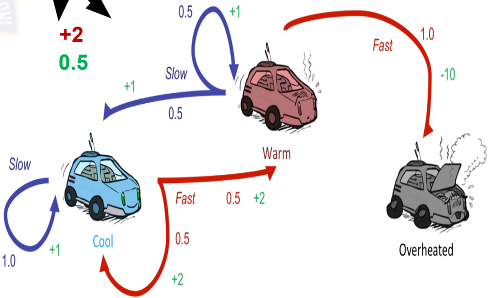
Policy Iteration (using iterative policy evaluation) for estimating $\pi \approx \pi_*$

1. Initialization
   $V(s) \in \mathbb{R}$ and $\pi(s) \in \mathcal{A}(s)$ arbitrarily for all $s \in \mathcal{S}$

$\gamma = 0.9$
$\theta = 0.01$

2. Policy Evaluation
   Loop:
   $\Delta \leftarrow 0$
   Loop for each $s \in \mathcal{S}$:
   $v \leftarrow V(s)$
   $V(s) \leftarrow \sum_{s',r} p(s',r|s,\pi(s))\big[r + \gamma V(s')\big]$
   $\Delta \leftarrow \max(\Delta, |v - V(s)|)$
   until $\Delta < \theta$ (a small positive number determining the accuracy of estimation)

3. Policy Improvement
   policy-stable $\leftarrow$ *true*
   For each $s \in \mathcal{S}$:
   old-action $\leftarrow \pi(s)$
   $\pi(s) \leftarrow \arg\max_a \sum_{s',r} p(s',r|s,a)\big[r + \gamma V(s')\big]$
   If *old-action* $\neq \pi(s)$, then *policy-stable* $\leftarrow$ *false*
   If *policy-stable*, then stop and return $V \approx v_*$ and $\pi \approx \pi_*$; else go to 2



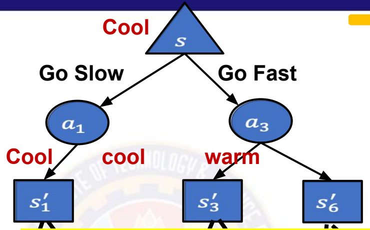
π_0

State	Best Action
Cool	Slow
Warm	Slow
Overheated	-

We will use asynchronous for updates in this example. Note that even though above backup diagram depicts all possible transition, this algorithm computes value from only one of the action as given by the initialized policy at the start of every iteration!!!

Policy Evaluation Starts – Policy Iteration

V_0	0	0	0
V_1	1	1.45	0
V_2			



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$\gamma = 0.9$
$\theta = 0.01$

2. Policy Evaluation
   Loop:
   $\Delta \leftarrow 0$
   Loop for each $s \in \mathcal{S}$:
   $v \leftarrow V(s)$
   $V(s) \leftarrow \sum_{s',r} p(s',r|s,\pi(s))\left[r + \gamma V(s')\right]$
   $\Delta \leftarrow \max(\Delta, |v - V(s)|)$
   until $\Delta < \theta$ (a small positive number determining the accuracy of estimation)

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   policy-stable $\leftarrow$ *true*
   For each $s \in \mathcal{S}$:
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   $\pi(s) \leftarrow \operatorname{argmax}_a \sum_{s',r} p(s',r|s,a)\left[r + \gamma V(s')\right]$
   If *old-action* $\neq \pi(s)$, then *policy-stable* $\leftarrow$ *false*
   If *policy-stable*, then stop and return $V \approx v_*$ and $\pi \approx \pi_*$; else go to 2

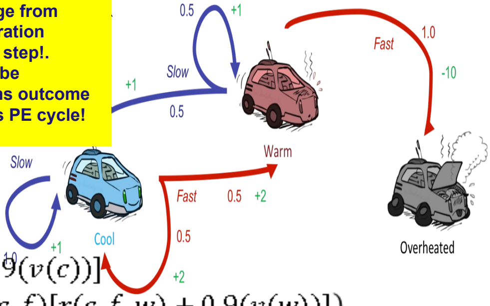
This slide show what change from Value iteration to Policy iteration algorithm in the Evaluation step!. Since only action needs to be assessed, MAX of all actions outcome is no longer required in this PE cycle!

- The value uses the initialized policy to select only one action GREEDILY. Hence below cancelled equations differentiated PI vs VI

$$V(cool) = \cancel{\max} \begin{cases} slow: 1 \\ \cancel{fast: 2} \end{cases}$$

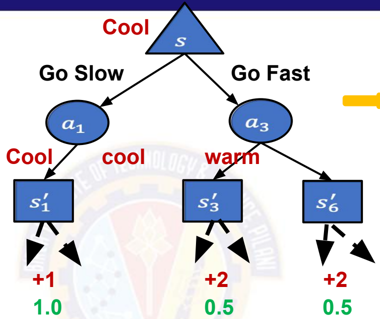
$$V(cool) = \cancel{\max} \begin{cases} 1 * [1 + 0.9(0)] \\ \cancel{(0.5 * [2 + 0.9(0)]) + (0.5[2 + 0.9(0)])} \end{cases}$$

$$V(cool) = \cancel{\max} \begin{cases} P(c|c,s)[r(c,s,c) + 0.9(v(c))] \\ \cancel{(P(c|c,f)[r(c,f,c) + 0.9(v(c))]) + (P(w|c,f)[r(c,f,w) + 0.9(v(w))])} \end{cases}$$



Policy Evaluation Completes – Policy Iteration

V_0	0	0	0
V_1	**1**	1.45	0
V_2	**1.9**	**2.51**	0
V_{225}	~10	~ 10	0

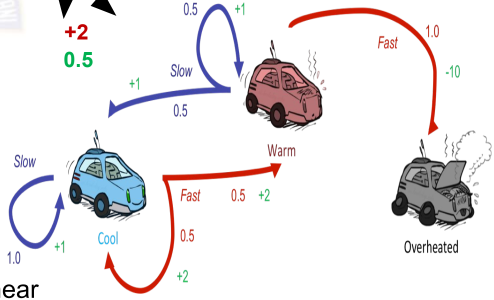


Policy Iteration (using iterative policy evaluation) for estimating $\pi \approx \pi_*$

1. Initialization
   $V(s) \in \mathbb{R}$ and $\pi(s) \in \mathcal{A}(s)$ arbitrarily for all $s \in \mathcal{S}$
2. Policy Evaluation
   Loop:
   $\Delta \leftarrow 0$
   Loop for each $s \in \mathcal{S}$:
   $v \leftarrow V(s)$
   $V(s) \leftarrow \sum_{s',r} p(s',r|s,\pi(s))[r + \gamma V(s')]$
   $\Delta \leftarrow \max(\Delta, |v - V(s)|)$
   until $\Delta < \theta$ (a small positive number determining the accuracy of estimation)
3. Policy Improvement
   policy-stable $\leftarrow$ *true*
   For each $s \in \mathcal{S}$:
   old-action $\leftarrow \pi(s)$
   $\pi(s) \leftarrow \operatorname{argmax}_a \sum_{s',r} p(s',r|s,a)[r + \gamma V(s')]$
   If *old-action* $\neq \pi(s)$, then *policy-stable* $\leftarrow$ *false*
   If *policy-stable*, then stop and return $V \approx v_*$ and $\pi \approx \pi_*$; else go to 2

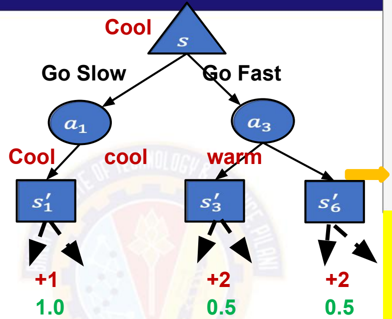
$\gamma = 0.9$
$\theta = 0.01$

- In PE, either synchronous or asynchronous value updates can be followed
- If Threshold of 0.01 is applied then approx., at 46th iteration algorithm converges
- Assume that Asynchronous method is used & Thresholding is not applied. In this case at approx., near 225th iteration values converge



Policy Improvement – Policy Iteration

V_0	0	0	0
V_1	1	1.45	0
V_2	1.9	2.51	0
V_{225}	~10	~ 10	0



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   Loop:
   $\Delta \leftarrow 0$
   Loop for each $s \in \mathcal{S}$:
   $v \leftarrow V(s)$
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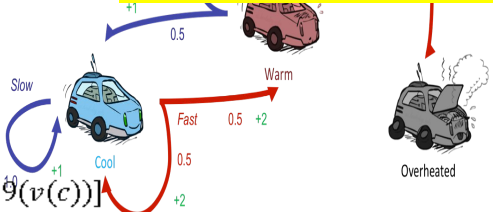
   If *policy-stable*, then stop and return $V \approx v_*$ and $\pi \approx \pi_*$; else go to 2

After PE is done, PI starts where the last value from PE is used to find the next best action. In this PI phase all actions are analysed and best action that MAXimises the value is chosen to update the policy

$$V(cool) = \max\begin{cases} slow: 10 \\ fast: 11 \end{cases}$$

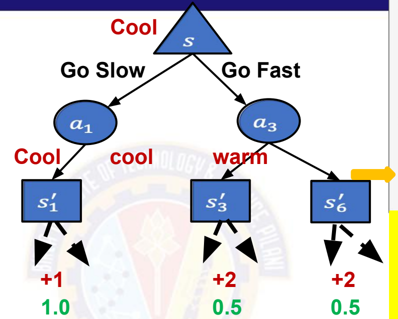
$$V(cool) = \max\begin{cases} 1 * [1 + 0.9(10)] \\ (0.5 * [2 + 0.9(10)]) + (0.5[2 + 0.9(10)]) \end{cases}$$

$$V(cool) = \max\begin{cases} P(c \mid c, s)[r(c,s,c) + 0.9(v(c))] \\ (P(c \mid c, f)[r(c,f,c) + 0.9(v(c))]) + (P(w \mid c, f)[r(c,f,w) + 0.9(v(w))]) \end{cases}$$



Policy Improvement – Policy Iteration

V_0	0	0	0
	……..		
V_1	1	1.45	0
	………..		
V_2	1.9	2.51	0
	………..		
V_{225}	~10	~ 10	0



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$\theta = 0.01$

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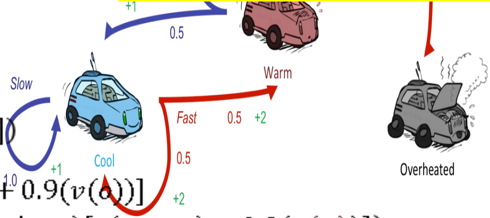


$$V(warm) = \max \begin{cases} fast: -10 \\ slow: 10 \end{cases}$$

$$V(warm) = \max \begin{cases} 1 * [-10 + 0.9(0)] \\ (0.5 * [1 + 0.9(10)]) + (0.5[1 + 0.9(10)]) \end{cases}$$

$$V(warm) = \max \begin{cases} P(o|w,f)[r(w,f,o) + 0.9(v(o))] \\ (P(c|w,s)[r(w,s,c) + 0.9(v(c))]) + (P(w|w,s)[r(w,s,w) + 0.9(v(w))]) \end{cases}$$

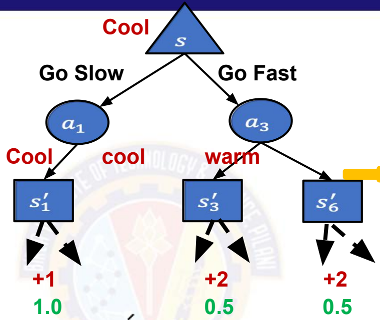
Policy Improvement Completes – Policy Iteration

V_0	0	0	0
V_1	1	1.45	0
V_2	1.9	2.51	0
V_{225}	~10	~ 10	0

$$V(warm) = \max\begin{cases} fast: -10 \\ slow: 10 \end{cases}$$

π_1

State	Best Action
Cool	Fast
Warm	Slow
Overheated	-



$$V(cool) = \max\begin{cases} slow: 10 \\ fast: 11 \end{cases}$$

Policy Iteration (using iterative policy evaluation) for estimating $\pi \approx \pi_*$

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   - If *old-action* $\neq \pi(s)$, then *policy-stable* $\leftarrow$ *false*

   If *policy-stable*, then stop and return $V \approx v_*$ and $\pi \approx \pi_*$; else go to 2

$\gamma = 0.9$
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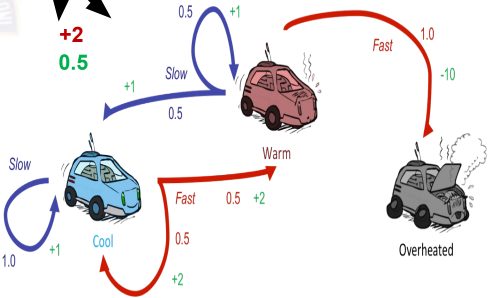


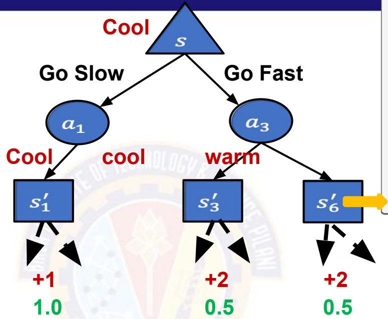
Continue another cycle of PE□PI – Policy Iteration

V_0: 10 10 0

....

V_1:

V_2:



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   $V(s) \in \mathbb{R}$ and $\pi(s) \in \mathcal{A}(s)$ arbitrarily for all $s \in \mathcal{S}$

$\gamma = 0.9$
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2. Policy Evaluation
   Loop:
   $\Delta \leftarrow 0$
   Loop for each $s \in \mathcal{S}$:
   $v \leftarrow V(s)$
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   For each $s \in \mathcal{S}$:
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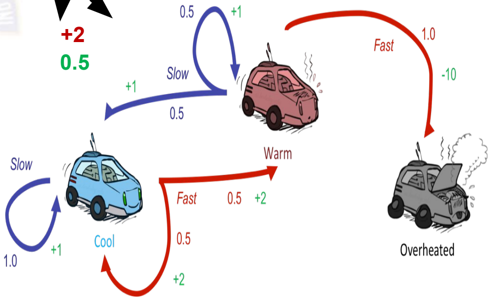


π_1

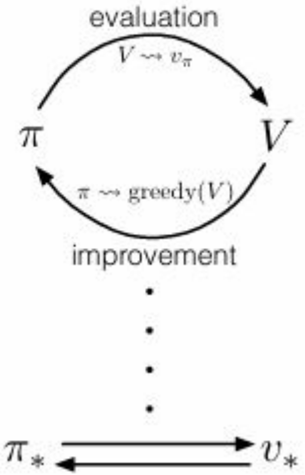
State	Best Action
Cool	Fast
Warm	Slow
Overheated	-

After this cycle of algorithm converges with values :
V(cool) = 15.5
V(Warm) = 14.5

Dynamic Programming - Policy Iteration

1 Observation

- Policy iteration consists of two simultaneous, interacting processes, one making the value function consistent with the current policy (policy evaluation), and the other making the policy greedy with respect to the current value function (policy improvement).
- In policy iteration, these two processes alternate, each completing before the other begins, **but this is not really necessary.**



Solution : GPI Generalized Policy Iteration

- Generalized policy iteration (GPI) to refer to the general idea of letting policy-evaluation and policy improvement processes interact, independent of the granularity and other details of the two processes. Almost all reinforcement learning methods are well described as GPI.

Idea (Do this as an exercise for given problem)

- In one cycle perform only one iteration of Policy Evaluation (ie.,Value Updates) without waiting for convergence immediately follow by Policy Improvement.

Table of Contents

1. MDP
2. DP
3. MC
4. TD (to be updated on or after session 9)

Problem Definition

Race Car Laps

In a modified Race car, there is a training track where agent is given a known (sometimes incomplete) map and risk factors(dynamics) may not be known in advance. Agent knows exactly where it is but its actions impact on the engine state change is difficult to predict (UNKNOWN MODEL).

With partial knowledge, the agent can compute optimal policies by trying through random experiments (Sampling). Car shuts down when its engine get overheated (optional terminal state) or Car stops once the predefined no.of.laps are completed.

Key Characteristics:

- The agent must act under **uncertainty** —
  percepts are local, and the agent must reason about hidden dangers
- Assume there are fuel constraints or/and No.of.Laps to complete!
- At every time step **there is no way clear indication of Impact of next** potential engine temperature depending on the current engine state. The entire past changes might lead to any possible state change. Hence **immediate reward is not always available**.

MARKOV DECISION PROCESS FORMALIZATION (Reduced)

1. States (S)
   - S = (Cool(**C**), Warm(**W**), Overheated(**O**)) #Engine Temperature
   - Finite State MDP
2. Actions (A)
   - A = {Go Slow(**S**), Go Fast(**F**)}

3. Rewards (R) and/or State Transition Probabilities (P) may be non-stationary

4. Policy still needs estimation of V(S) , Q(S,A) i.e., MDP components but its is not a necessity that the process is expected to follow a Markov Property!! ☺

Monte Carlo Methods

1. Given
   - S, A, optionally R(s, a, s`)
2. To find
   - Policy π
3. Solution
   - Sample Episodes
   - Compute estimates of the value by discounted returns
     - ✔ Simple average
     - ✔ Weighted average
     - ✔ Normalized weighted average
4. Idea
   - On Policy :
     - Generate Sample/Experience □ Evaluate using Return □ Improve
   - Off Policy :
     - Refer to Sample/Experience generated by another reference(behavior) policy □ Evaluate using Return □ Improve

Monte Carlo Method

On Policy Control Algorithm

Monte Carlo

Q_0	0	0	0	0	0	0
Q_1						
Q_2						

On-policy first-visit MC control (for ε-soft policies), estimates $\pi \approx \pi_*$

Algorithm parameter: small $\varepsilon > 0$

Initialize:
- $\pi \leftarrow$ an arbitrary ε-soft policy
- $Q(s,a) \in \mathbb{R}$ (arbitrarily), for all $s \in \mathcal{S}$, $a \in \mathcal{A}(s)$
- $Returns(s,a) \leftarrow$ empty list, for all $s \in \mathcal{S}$, $a \in \mathcal{A}(s)$

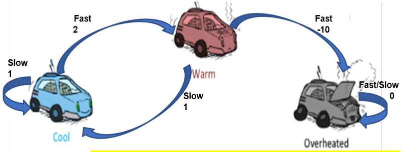
Repeat forever (for each episode):
- Generate an episode following π: $S_0, A_0, R_1, \ldots, S_{T-1}, A_{T-1}, R_T$
- $G \leftarrow 0$
- Loop for each step of episode, $t = T-1, T-2, \ldots, 0$:
  - $G \leftarrow \gamma G + R_{t+1}$
  - Unless the pair S_t, A_t appears in $S_0, A_0, S_1, A_1 \ldots, S_{t-1}, A_{t-1}$:
    - Append G to $Returns(S_t, A_t)$
    - $Q(S_t, A_t) \leftarrow \text{average}(Returns(S_t, A_t))$
    - $A^* \leftarrow \arg\max_a Q(S_t, a)$ (with ties broken arbitrarily)
    - For all $a \in \mathcal{A}(S_t)$:

$$\pi(a|S_t) \leftarrow \begin{cases} 1 - \varepsilon + \varepsilon/|\mathcal{A}(S_t)| & \text{if } a = A^* \\ \varepsilon/|\mathcal{A}(S_t)| & \text{if } a \neq A^* \end{cases}$$

□ = 0.2
γ = 0.6

π_0

State	Best Action	P(F)	P(S)
Cool	Slow	0.4	0.6
Warm	Fast	0.6	0.4
Overheated	-	-	-



First Visit Policy Evaluation

Monte Carlo

Q_0	0	0	0	0	0	0
Q_1						
Q_2						

Episode 1: C, F, 2, W, S, 1, C, F, 2, W, F, -10, O

G=0

On-policy first-visit MC control (for ε-soft policies), estimates $\pi \approx \pi_*$

Algorithm parameter: small $\varepsilon > 0$

Initialize:

$\pi \leftarrow$ an arbitrary ε-soft policy

$Q(s,a) \in \mathbb{R}$ (arbitrarily), for all $s \in \mathcal{S}$, $a \in \mathcal{A}(s)$

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$G \leftarrow \gamma G + R_{t+1}$

Unless the pair S_t, A_t appears in $S_0, A_0, S_1, A_1 \ldots, S_{t-1}, A_{t-1}$:

Append G to $Returns(S_t, A_t)$

$Q(S_t, A_t) \leftarrow \text{average}(Returns(S_t, A_t))$

$A^* \leftarrow \arg\max_a Q(S_t, a)$ (with ties broken arbitrarily)

For all $a \in \mathcal{A}(S_t)$:

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□ = 0.2

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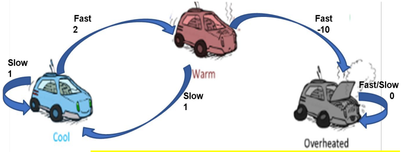
π_0

State	Best Action	P(F)	P(S)
Cool	Slow	0.4	0.6
Warm	Fast	0.6	0.4
Overheated	-	-	-



First Visit Policy Evaluation

Monte Carlo

Q_0	0	0	0	0	0	0
Q_1	-4 ,1.16	0	-10	-1.4	0	0
Q_2						

Episode 1: C, F, 2, W, S, 1, C, F, 2, W, F, -10, O
G=0

t=3: G_3 = -10
t=2: G_2 = 2 + (0.6)(-10) = -4 #Not used in First Visit
t=1: G_1 = 1+ (0.6*2) + (0.6*0.6*-10) = -1.4
t=0: G_0 = 2+ (0.6*1) + (0.6*0.6*2) + (0.6*0.6*0.6-10) = 1.16

On-policy first-visit MC control (for ε-soft policies), estimates $\pi \approx \pi_*$

Algorithm parameter: small $\varepsilon > 0$
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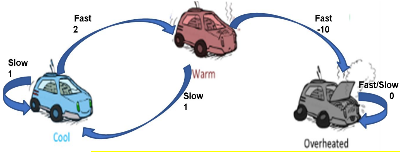

First Visit Policy Evaluation

Monte Carlo

Q_0	0	0	0	0	0	0
Q_1	~~-4~~,1.16	0	-10	-1.4	0	0
Q_2						

Episode 1: C, F, 2, W, S, 1, C, F, 2, W, F, -10, O

G=0

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t=0: G_0 = 2+ (0.6*1) + (0.6*0.6*2) + (0.6*0.6*0.6-10) = 1.16

Episode 1:

t=3: Q(W, F) = -10
t=2: Q(C, F) = -4
t=1: Q(W, S) = -1.4
t=0: Q(C, F) = 1.16

On-policy first-visit MC control (for ε-soft policies), estimates $\pi \approx \pi_*$

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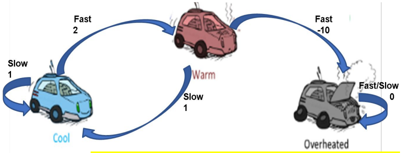
      $A^* \leftarrow \arg\max_a Q(S_t, a)$ (with ties broken arbitrarily)
      For all $a \in \mathcal{A}(S_t)$:
        $$\pi(a|S_t) \leftarrow \begin{cases} 1 - \varepsilon + \varepsilon/|\mathcal{A}(S_t)| & \text{if } a = A^* \\ \varepsilon/|\mathcal{A}(S_t)| & \text{if } a \neq A^* \end{cases}$$

ε = 0.2
γ = 0.6

π_0

State	Best Action	P(F)	P(S)
Cool	Slow	0.4	0.6
Warm	Fast	0.6	0.4
Overheated	-	-	-



First Visit Policy Evaluation

Monte Carlo

Q_0	0	0	0	0	0	0
Q_1	1.16	0	-10	-1.4	0	0
Q_2						

Episode 1: C, F, 2, W, S, 1, C, F, 2, W, F, -10, O
G=0

t=3: G_3 = -10
t=2: G_2 = 2 + (0.6)(-10) = -4 #Not used in First Visit
t=1: G_1 = 1+ (0.6*2) + (0.6*0.6*-10) = -1.4
t=0: G_0 = 2+ (0.6*1) + (0.6*0.6*2) + (0.6*0.6*0.6-10) = 1.16

Episode 1:

t=3: Q(W, F) = -10
t=2: Q(C, F) = -4
t=1: Q(W, S) = -1.4
t=0: Q(C, F) = 1.16

On-policy first-visit MC control (for ε-soft policies), estimates $\pi \approx \pi_*$

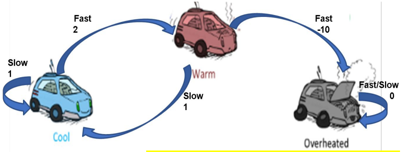
Algorithm parameter: small $\varepsilon > 0$
Initialize:
$\pi \leftarrow$ an arbitrary ε-soft policy
$Q(s,a) \in \mathbb{R}$ (arbitrarily), for all $s \in \mathcal{S}$, $a \in \mathcal{A}(s)$
$Returns(s,a) \leftarrow$ empty list, for all $s \in \mathcal{S}$, $a \in \mathcal{A}(s)$

Repeat forever (for each episode):
Generate an episode following π: $S_0, A_0, R_1, \ldots, S_{T-1}, A_{T-1}, R_T$
$G \leftarrow 0$
Loop for each step of episode, $t = T-1, T-2, \ldots, 0$:
$G \leftarrow \gamma G + R_{t+1}$
Unless the pair S_t, A_t appears in $S_0, A_0, S_1, A_1 \ldots, S_{t-1}, A_{t-1}$:
Append G to $Returns(S_t, A_t)$
$Q(S_t, A_t) \leftarrow \text{average}(Returns(S_t, A_t))$
$A^* \leftarrow \arg\max_a Q(S_t, a)$ (with ties broken arbitrarily)
For all $a \in \mathcal{A}(S_t)$:
$$\pi(a|S_t) \leftarrow \begin{cases} 1 - \varepsilon + \varepsilon/|\mathcal{A}(S_t)| & \text{if } a = A^* \\ \varepsilon/|\mathcal{A}(S_t)| & \text{if } a \neq A^* \end{cases}$$

□ = 0.2
γ = 0.6

π_1

State	Best Action	P(F)	P(S)
Cool	Slow	0.2	0.8
Warm	Slow	0.2	0.8
Overheated	-	-	-



Policy Improvement

Monte Carlo

	Cool FAST	Cool SLOW	Warm FAST	Warm SLOW	Overheated FAST	Overheated SLOW
Q_0	0	0	0	0	0	0
Q_1	**1.16**	0	**-10**	**-1.4**	0	0
Q_2						

On-policy first-visit MC control (for ε-soft policies), estimates $\pi \approx \pi_*$

Algorithm parameter: small $\varepsilon > 0$
Initialize:
  $\pi \leftarrow$ an arbitrary ε-soft policy
  $Q(s,a) \in \mathbb{R}$ (arbitrarily), for all $s \in \mathcal{S}$, $a \in \mathcal{A}(s)$
  $Returns(s,a) \leftarrow$ empty list, for all $s \in \mathcal{S}$, $a \in \mathcal{A}(s)$

Repeat forever (for each episode):
  Generate an episode following π: $S_0, A_0, R_1, \ldots, S_{T-1}, A_{T-1}, R_T$
  $G \leftarrow 0$
  Loop for each step of episode, $t = T-1, T-2, \ldots, 0$:
    $G \leftarrow \gamma G + R_{t+1}$
    Unless the pair S_t, A_t appears in $S_0, A_0, S_1, A_1 \ldots, S_{t-1}, A_{t-1}$:
      Append G to $Returns(S_t, A_t)$
      $Q(S_t, A_t) \leftarrow \text{average}(Returns(S_t, A_t))$
      $A^* \leftarrow \arg\max_a Q(S_t, a)$ (with ties broken arbitrarily)
      For all $a \in \mathcal{A}(S_t)$:
$$\pi(a|S_t) \leftarrow \begin{cases} 1 - \varepsilon + \varepsilon/|\mathcal{A}(S_t)| & \text{if } a = A^* \\ \varepsilon/|\mathcal{A}(S_t)| & \text{if } a \neq A^* \end{cases}$$

□ = 0.2
γ = 0.6

π_1

State	Best Action	P(F)	P(S)
Cool	Slow	0.2	0.8
Warm	Slow	0.2	0.8
Overheated	-	-	-

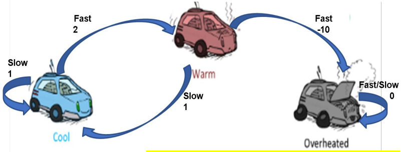
Episode 1: C, F, 2, W, S, 1, C, F, 2, W, F, -10, O
G=0

t=3: G_3 = -10
t=2: G_2 = 2 + (0.6)(-10) = -4 #Not used in First Visit
t=1: G_1 = 1+ (0.6*2) + (0.6*0.6*-10) = -1.4
t=0: G_0 = 2+ (0.6*1) + (0.6*0.6*2) + (0.6*0.6*0.6-10) = 1.16

Episode 1:

t=3: Q(W, F) = -10
t=2: Q(C, F) = -4
t=1: Q(W, S) = -1.4
t=0: Q(C, F) = 1.16

Episode 2: C, S, 1, C, F, 2, W, F, -10, O
G=0
t=2: G_2 = -10
t=1: G_1 = 2 + (0.6)(-10) = -4
t=0: G_0 = 1+ (0.6*2) + (0.6*0.6*-10) = -1.4



Policy Evaluation

Monte Carlo

	FAST	SLOW	FAST	SLOW	FAST	SLOW
Q_0	0	0	0	0	0	0
Q_1	1.16	0	-10	-1.4	0	0
Q_2	-1.42	-1.4	-10	0	0	0

On-policy first-visit MC control (for ε-soft policies), estimates $\pi \approx \pi_*$

Algorithm parameter: small $\varepsilon > 0$
Initialize:
  $\pi \leftarrow$ an arbitrary ε-soft policy
  $Q(s,a) \in \mathbb{R}$ (arbitrarily), for all $s \in \mathcal{S}$, $a \in \mathcal{A}(s)$
  $Returns(s,a) \leftarrow$ empty list, for all $s \in \mathcal{S}$, $a \in \mathcal{A}(s)$

Repeat forever (for each episode):
  Generate an episode following π: $S_0, A_0, R_1, \ldots, S_{T-1}, A_{T-1}, R_T$
  $G \leftarrow 0$
  Loop for each step of episode, $t = T-1, T-2, \ldots, 0$:
    $G \leftarrow \gamma G + R_{t+1}$
    Unless the pair S_t, A_t appears in $S_0, A_0, S_1, A_1 \ldots, S_{t-1}, A_{t-1}$:
      Append G to $Returns(S_t, A_t)$
      $Q(S_t, A_t) \leftarrow \text{average}(Returns(S_t, A_t))$
      $A^* \leftarrow \arg\max_a Q(S_t, a)$ (with ties broken arbitrarily)
      For all $a \in \mathcal{A}(S_t)$:

$$\pi(a|S_t) \leftarrow \begin{cases} 1 - \varepsilon + \varepsilon/|\mathcal{A}(S_t)| & \text{if } a = A^* \\ \varepsilon/|\mathcal{A}(S_t)| & \text{if } a \neq A^* \end{cases}$$

□ = 0.2
γ = 0.6

π_1

State	Best Action	P(F)	P(S)
Cool	Slow	0.2	0.8
Warm	Slow	0.2	0.8
Overheated	-	-	-

Episode 1: C, F, 2, W, S, 1, C, F, 2, W, F, -10, O
G=0

t=3: G_3 = -10
t=2: G_2 = 2 + (0.6)(-10) = -4 #Not used in First Visit
t=1: G_1 = 1+ (0.6*2) + (0.6*0.6*-10) = -1.4
t=0: G_0 = 2+ (0.6*1) + (0.6*0.6*2) + (0.6*0.6*0.6-10) = 1.16

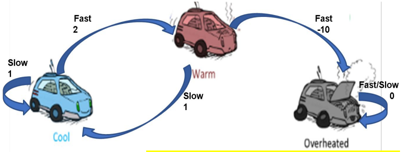
Episode 1:

t=3: Q(W, F) = -10
t=2: Q(C, F) = -4
t=1: Q(W, S) = -1.4
t=0: Q(C, F) = 1.16

Episode 2: C, S, 1, C, F, 2, W, F, -10, O
G=0
t=2: G_2 = -10
t=1: G_1 = 2 + (0.6)(-10) = -4
t=0: G_0 = 1+ (0.6*2) + (0.6*0.6*-10) = -1.4

Episode 2:

t=2: Q(W, F) = -10
t=1: Q(C, F) = avg(1.16,-4) = -1.42
t=0: Q(C, S) = -1.4



Policy PE & PI

Monte Carlo

Q_0	0	0	0	0	0	0
Q_1	1.16	0	-10	-1.4	0	0
Q_2	-1.42	-1.4	-10	0	0	0

On-policy first-visit MC control (for ε-soft policies), estimates $\pi \approx \pi_*$

Algorithm parameter: small $\varepsilon > 0$

Initialize:

$\pi \leftarrow$ an arbitrary ε-soft policy

$Q(s,a) \in \mathbb{R}$ (arbitrarily), for all $s \in \mathcal{S}$, $a \in \mathcal{A}(s)$

$Returns(s,a) \leftarrow$ empty list, for all $s \in \mathcal{S}$, $a \in \mathcal{A}(s)$

Repeat forever (for each episode):

Generate an episode following π: $S_0, A_0, R_1, \ldots, S_{T-1}, A_{T-1}, R_T$

$G \leftarrow 0$

Loop for each step of episode, $t = T-1, T-2, \ldots, 0$:

$G \leftarrow \gamma G + R_{t+1}$

Unless the pair S_t, A_t appears in $S_0, A_0, S_1, A_1 \ldots, S_{t-1}, A_{t-1}$:

Append G to $Returns(S_t, A_t)$

$Q(S_t, A_t) \leftarrow \text{average}(Returns(S_t, A_t))$

$A^* \leftarrow \text{argmax}_a Q(S_t, a)$ (with ties broken arbitrarily)

For all $a \in \mathcal{A}(S_t)$:

$$\pi(a|S_t) \leftarrow \begin{cases} 1 - \varepsilon + \varepsilon/|\mathcal{A}(S_t)| & \text{if } a = A^* \\ \varepsilon/|\mathcal{A}(S_t)| & \text{if } a \neq A^* \end{cases}$$

□ = 0.2

γ = 0.6

π_1

State	Best Action	P(F)	P(S)
Cool	Slow	0.2	0.8
Warm	Slow	0.2	0.8
Overheated	-	-	-

Episode 3: C, S, 1, C, S, 1, C, F, 2, W, F, -10, O

G=0

t=3: G_3 = -10

t=2: G_2 = 2 + (0.6)(-10) = -4

t=1: G_1 = 1+ (0.6*2) + (0.6*0.6*-10) = -1.4

t=0: G_0 = 1+ (0.6*1) + (0.6*0.6*2) + (0.6*0.6*0.6-10) = 0.16

Episode 3:

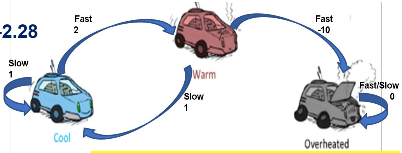
t=3: Q(W, F) = -10

t=2: Q(C, F) = avg(-4,1.16,-4) = -2.28

t=1: Q(C, S) = -1.4

t=0: Q(C, S) = avg(0.16,-1.4)

= -0.62



First Visit Policy Evaluation

Monte Carlo

Q_0	0	0	0	0	0	0
Q_1	1.16	0	-10	-1.4	0	0
Q_2	-1.42	-1.4	-10	0	0	0
Q_3	-2.28	-0.62	-10	0	0	0

On-policy first-visit MC control (for ε-soft policies), estimates $\pi \approx \pi_*$

Algorithm parameter: small $\varepsilon > 0$
Initialize:
  $\pi \leftarrow$ an arbitrary ε-soft policy
  $Q(s,a) \in \mathbb{R}$ (arbitrarily), for all $s \in \mathcal{S}$, $a \in \mathcal{A}(s)$
  $Returns(s,a) \leftarrow$ empty list, for all $s \in \mathcal{S}$, $a \in \mathcal{A}(s)$

Repeat forever (for each episode):
  Generate an episode following π: $S_0, A_0, R_1, \ldots, S_{T-1}, A_{T-1}, R_T$
  $G \leftarrow 0$
  Loop for each step of episode, $t = T-1, T-2, \ldots, 0$:
    $G \leftarrow \gamma G + R_{t+1}$
    Unless the pair S_t, A_t appears in $S_0, A_0, S_1, A_1 \ldots, S_{t-1}, A_{t-1}$:
      Append G to $Returns(S_t, A_t)$
      $Q(S_t, A_t) \leftarrow \text{average}(Returns(S_t, A_t))$
      $A^* \leftarrow \arg\max_a Q(S_t, a)$ (with ties broken arbitrarily)
      For all $a \in \mathcal{A}(S_t)$:
$$\pi(a|S_t) \leftarrow \begin{cases} 1 - \varepsilon + \varepsilon/|\mathcal{A}(S_t)| & \text{if } a = A^* \\ \varepsilon/|\mathcal{A}(S_t)| & \text{if } a \neq A^* \end{cases}$$

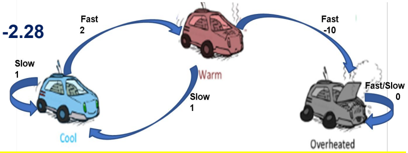
□ = 0.2
γ = 0.6

π_2

State	Best Action	P(F)	P(S)
Cool	Slow	0.2	0.8
Warm	Slow	0.2	0.8
Overheated	-	-	-

Episode 3:

t=3: Q(W, F) = -10
t=2: Q(C, F) = avg(-4,1.16,-4) = -2.28
t=1: Q(C, S) = -1.4
t=0: Q(C, S) = avg(0.16,-1.4)
= -0.62



After PE PI no change in π.
Policy converges towards safer actions

Monte Carlo Methods

1. Given
   - S, A, optionally R(s, a, s`)
2. To find
   - Policy π . This is referred to as **TARGET** policy
3. Solution
   - Sample Episodes
   - Compute estimates of the value by discounted returns
     - ✔ Simple average
     - ✔ Weighted average
     - ✔ Normalized weighted average
4. Idea
   - On Policy :
     - Generate Sample/Experience □ Evaluate using Return □ Improve
   - Off Policy :
     - Refer to Sample/Experience generated by another reference(**BEHAVIOUR**) policy □ Evaluate using Return □ Improve

Monte Carlo Method

Off Policy Control Algorithm

Monte Carlo

Q	0	0	0	0	0	0
π						
C						

Off-policy MC control, for estimating $\pi \approx \pi_*$

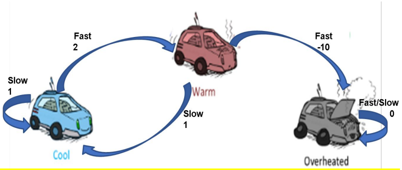
□ = 0.2
γ = 0.6

Initialize, for all $s \in \mathcal{S}$, $a \in \mathcal{A}(s)$:
  $Q(s,a) \leftarrow$ arbitrary
  $C(s,a) \leftarrow 0$
  $\pi(s) \leftarrow \operatorname{argmax}_a Q(S_t, a)$ (with ties broken consistently)

Repeat forever:
  $b \leftarrow$ any soft policy
  Generate an episode using b:
    $S_0, A_0, R_1, \ldots, S_{T-1}, A_{T-1}, R_T, S_T$
  $G \leftarrow 0$
  $W \leftarrow 1$
  For $t = T-1, T-2, \ldots$ down to 0:
    $G \leftarrow \gamma G + R_{t+1}$
    $C(S_t, A_t) \leftarrow C(S_t, A_t) + W$
    $Q(S_t, A_t) \leftarrow Q(S_t, A_t) + \frac{W}{C(S_t, A_t)} [G - Q(S_t, A_t)]$
    $\pi(S_t) \leftarrow \operatorname{argmax}_a Q(S_t, a)$ (with ties broken consistently)
    If $A_t \neq \pi(S_t)$ then exit For loop
    $W \leftarrow W \frac{1}{b(A_t|S_t)}$

b

State	P(F)	P(S)
Cool	0.4	0.6
Warm	0.6	0.4
Overheated	-	-



Assume deterministic Target policy. Hence $\pi(A_t \mid S_t) = 1$
But the behavior policy (b) is Stochastic as defined

Monte Carlo

	FAST	SLOW	FAST	SLOW	FAST	SLOW
Q	0	0	0	0	0	0
π		✔	✔			
C	0	0	0	0	0	0

Ep	
G	
W	
t	
A_t	
$\pi(S_t)$	

Off-policy MC control, for estimating $\pi \approx \pi_*$

□ = 0.2

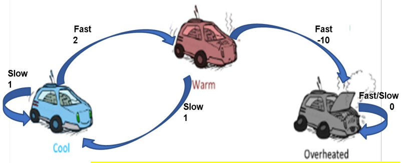
γ = 0.6

Initialize, for all $s \in \mathcal{S}$, $a \in \mathcal{A}(s)$:
  $Q(s,a) \leftarrow$ arbitrary
  $C(s,a) \leftarrow 0$
  $\pi(s) \leftarrow \operatorname{argmax}_a Q(S_t,a)$ (with ties broken consistently)

Repeat forever:
  $b \leftarrow$ any soft policy
  Generate an episode using b:
    $S_0, A_0, R_1, \ldots, S_{T-1}, A_{T-1}, R_T, S_T$
  $G \leftarrow 0$
  $W \leftarrow 1$
  For $t = T-1, T-2, \ldots$ down to 0:
    $G \leftarrow \gamma G + R_{t+1}$
    $C(S_t,A_t) \leftarrow C(S_t,A_t) + W$
    $Q(S_t,A_t) \leftarrow Q(S_t,A_t) + \frac{W}{C(S_t,A_t)}[G - Q(S_t,A_t)]$
    $\pi(S_t) \leftarrow \operatorname{argmax}_a Q(S_t,a)$ (with ties broken consistently)
    If $A_t \neq \pi(S_t)$ then exit For loop
    $W \leftarrow W \frac{1}{b(A_t|S_t)}$

b

State	P(F)	P(S)
Cool	0.4	0.6
Warm	0.6	0.4
Overheated	-	-



Every Visit Policy Evaluation

Monte Carlo

	Cool FAST	Cool SLOW	Warm FAST	Warm SLOW	Overheated FAST	Overheated SLOW
Q	0	0	0	0	0	0
π		✔	✔			
C	0	0	0	0	0	0

Ep	1
G	0
W	1
t	
A_t	
$\pi(S_t)$	

Episode 1: C, F, 2, W, S, 1, C, F, 2, W, F, -10, O

G=0

Off-policy MC control, for estimating $\pi \approx \pi_*$

☐ = 0.2

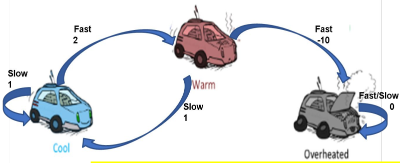
γ = 0.6

Initialize, for all $s \in \mathcal{S}$, $a \in \mathcal{A}(s)$:
- $Q(s,a) \leftarrow$ arbitrary
- $C(s,a) \leftarrow 0$
- $\pi(s) \leftarrow \arg\max_a Q(S_t,a)$ (with ties broken consistently)

Repeat forever:
- $b \leftarrow$ any soft policy
- Generate an episode using b:
  - $S_0, A_0, R_1, \ldots, S_{T-1}, A_{T-1}, R_T, S_T$
- $G \leftarrow 0$
- $W \leftarrow 1$
- For $t = T-1, T-2, \ldots$ down to 0:
  - $G \leftarrow \gamma G + R_{t+1}$
  - $C(S_t,A_t) \leftarrow C(S_t,A_t) + W$
  - $Q(S_t,A_t) \leftarrow Q(S_t,A_t) + \frac{W}{C(S_t,A_t)}[G - Q(S_t,A_t)]$
  - $\pi(S_t) \leftarrow \arg\max_a Q(S_t,a)$ (with ties broken consistently)
  - If $A_t \neq \pi(S_t)$ then exit For loop
  - $W \leftarrow W \frac{1}{b(A_t|S_t)}$

b

State	P(F)	P(S)
Cool	0.4	0.6
Warm	0.6	0.4
Overheated	-	-



First Visit Policy Evaluation

Monte Carlo

Q	0	0	0	0	0	0
π		✔	✔			
C	0	0	0	0	0	0

Ep	**1**
G	**0**
W	**1**
t	3
A_t	F
$\pi(S_t)$	F

Episode 1: C, F, 2, W, S, 1, C, F, 2, $\boxed{\text{W, F, -10, O}}$

G=0

t=3: G_3 = -10

Off-policy MC control, for estimating $\pi \approx \pi_*$

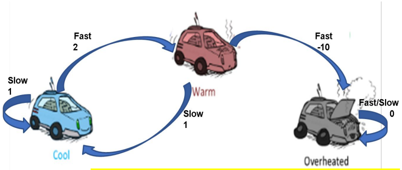
□ = 0.2
γ = 0.6

Initialize, for all $s \in \mathcal{S}$, $a \in \mathcal{A}(s)$:
$Q(s,a) \leftarrow$ arbitrary
$C(s,a) \leftarrow 0$
$\pi(s) \leftarrow \operatorname{argmax}_a Q(S_t,a)$ (with ties broken consistently)

Repeat forever:
$b \leftarrow$ any soft policy
Generate an episode using b:
$S_0, A_0, R_1, \ldots, S_{T-1}, A_{T-1}, R_T, S_T$
$G \leftarrow 0$
$W \leftarrow 1$
For $t = T-1, T-2, \ldots$ down to 0:
$G \leftarrow \gamma G + R_{t+1}$
$C(S_t,A_t) \leftarrow C(S_t,A_t) + W$
$Q(S_t,A_t) \leftarrow Q(S_t,A_t) + \frac{W}{C(S_t,A_t)}\left[G - Q(S_t,A_t)\right]$
$\pi(S_t) \leftarrow \operatorname{argmax}_a Q(S_t,a)$ (with ties broken consistently)
If $A_t \neq \pi(S_t)$ then exit For loop
$W \leftarrow W \frac{1}{b(A_t|S_t)}$

b

State	P(F)	P(S)
Cool	0.4	0.6
Warm	0.6	0.4
Overheated	-	-



First Visit Policy Evaluation

Monte Carlo

Q	0	0	0	0	0	0
π		✔	✔			
C	0	0	**1**	0	0	0

Ep	**1**
G	**-10**
W	**1**
t	3
A_t	F
$\pi(S_t)$	F

Episode 1: C, F, 2, W, S, 1, C, F, 2, W, F, -10, O

G=0

t=3: G_3 = -10

Off-policy MC control, for estimating $\pi \approx \pi_*$

□ **= 0.2**

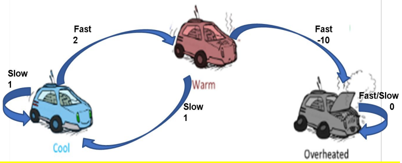
γ = 0.6

Initialize, for all $s \in \mathcal{S}$, $a \in \mathcal{A}(s)$:
- $Q(s,a) \leftarrow$ arbitrary
- $C(s,a) \leftarrow 0$
- $\pi(s) \leftarrow \operatorname{argmax}_a Q(S_t, a)$ (with ties broken consistently)

Repeat forever:
- $b \leftarrow$ any soft policy
- Generate an episode using b:
  - $S_0, A_0, R_1, \ldots, S_{T-1}, A_{T-1}, R_T, S_T$
- $G \leftarrow 0$
- $W \leftarrow 1$
- For $t = T-1, T-2, \ldots$ down to 0:
  - $G \leftarrow \gamma G + R_{t+1}$
  - $C(S_t, A_t) \leftarrow C(S_t, A_t) + W$
  - $Q(S_t, A_t) \leftarrow Q(S_t, A_t) + \frac{W}{C(S_t, A_t)}[G - Q(S_t, A_t)]$
  - $\pi(S_t) \leftarrow \operatorname{argmax}_a Q(S_t, a)$ (with ties broken consistently)
  - If $A_t \neq \pi(S_t)$ then exit For loop
  - $W \leftarrow W \frac{1}{b(A_t|S_t)}$

b

State	P(F)	P(S)
Cool	0.4	0.6
Warm	0.6	0.4
Overheated	-	-



Compute Return & Update the Count for Averaging

Monte Carlo

	FAST	SLOW	FAST	SLOW	FAST	SLOW
Q	0	0	-10	0	0	0
π		✔	✔			
C	0	0	1	0	0	0

Ep	1
G	-10
W	1
t	3
A_t	F
$\pi(S_t)$	F

Episode 1: C, F, 2, W, S, 1, C, F, 2, $\boxed{\text{W, F, -10, O}}$
G=0

t=3: G_3 = -10

Off-policy MC control, for estimating $\pi \approx \pi_*$

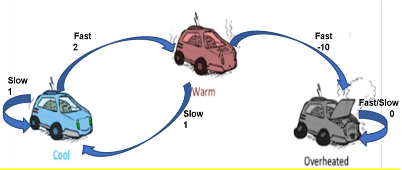
□ = 0.2
γ = 0.6

Initialize, for all $s \in \mathcal{S}$, $a \in \mathcal{A}(s)$:
- $Q(s,a) \leftarrow$ arbitrary
- $C(s,a) \leftarrow 0$
- $\pi(s) \leftarrow \operatorname{argmax}_a Q(S_t, a)$ (with ties broken consistently)

Repeat forever:
- $b \leftarrow$ any soft policy
- Generate an episode using b:
  - $S_0, A_0, R_1, \ldots, S_{T-1}, A_{T-1}, R_T, S_T$
- $G \leftarrow 0$
- $W \leftarrow 1$
- For $t = T-1, T-2, \ldots$ down to 0:
  - $G \leftarrow \gamma G + R_{t+1}$
  - $C(S_t, A_t) \leftarrow C(S_t, A_t) + W$
  - $Q(S_t, A_t) \leftarrow Q(S_t, A_t) + \frac{W}{C(S_t, A_t)}[G - Q(S_t, A_t)]$
  - $\pi(S_t) \leftarrow \operatorname{argmax}_a Q(S_t, a)$ (with ties broken consistently)
  - If $A_t \neq \pi(S_t)$ then exit For loop
  - $W \leftarrow W \frac{1}{b(A_t|S_t)}$

b

State	P(F)	P(S)
Cool	0.4	0.6
Warm	0.6	0.4
Overheated	-	-



Compute the action value Q(W,F) using incremental update form adapted to perform normalized weighted average based updates. This is the WEIGHTED IMPORTANCE SAMPLING METHOD, which bounds the variance due to the normalizing effect!

Monte Carlo

	FAST	SLOW	FAST	SLOW	FAST	SLOW
Q	0	0	**-10**	0	0	0
π		✔		✔		
C	0	0	1	0	0	0

Ep	**1**
G	**-10**
W	**1**
t	3
A_t	F
$\pi(S_t)$	S

Episode 1: C, F, 2, W, S, 1, C, F, 2, W, F, -10, O

G=0

t=3: G_3 = -10

Policy Improvement

Off-policy MC control, for estimating $\pi \approx \pi_*$

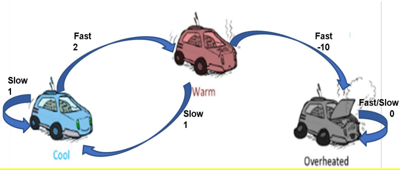
□ = 0.2
γ = 0.6

Initialize, for all $s \in \mathcal{S}$, $a \in \mathcal{A}(s)$:
  $Q(s,a) \leftarrow$ arbitrary
  $C(s,a) \leftarrow 0$
  $\pi(s) \leftarrow \text{argmax}_a Q(S_t, a)$ (with ties broken consistently)

Repeat forever:
  $b \leftarrow$ any soft policy
  Generate an episode using b:
    $S_0, A_0, R_1, \ldots, S_{T-1}, A_{T-1}, R_T, S_T$
  $G \leftarrow 0$
  $W \leftarrow 1$
  For $t = T-1, T-2, \ldots$ down to 0:
    $G \leftarrow \gamma G + R_{t+1}$
    $C(S_t, A_t) \leftarrow C(S_t, A_t) + W$
    $Q(S_t, A_t) \leftarrow Q(S_t, A_t) + \frac{W}{C(S_t, A_t)} [G - Q(S_t, A_t)]$
    $\pi(S_t) \leftarrow \text{argmax}_a Q(S_t, a)$ (with ties broken consistently)
    If $A_t \neq \pi(S_t)$ then exit For loop
    $W \leftarrow W \frac{1}{b(A_t|S_t)}$

b

State	P(F)	P(S)
Cool	0.4	0.6
Warm	0.6	0.4
Overheated	-	-



The best action is not the currently encountered action in the sample!

Monte Carlo

	FAST	SLOW	FAST	SLOW	FAST	SLOW
Q	0	0	**-10**	0	0	0
π		✔		✔		
C	0	0	1	0	0	0

Ep	**1**
G	**-10**
W	**1**
t	3
A_t	F
$\pi(S_t)$	S

Episode 1: C, F, 2, W, S, 1, C, F, 2, W, F, -10, O
G=0

t=3: G_3 = -10

Off-policy MC control, for estimating $\pi \approx \pi_*$

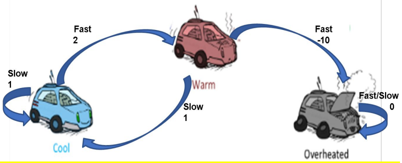
□ = 0.2
γ = 0.6

Initialize, for all $s \in \mathcal{S}$, $a \in \mathcal{A}(s)$:
  $Q(s,a) \leftarrow$ arbitrary
  $C(s,a) \leftarrow 0$
  $\pi(s) \leftarrow \arg\max_a Q(S_t, a)$ (with ties broken consistently)

Repeat forever:
  $b \leftarrow$ any soft policy
  Generate an episode using b:
    $S_0, A_0, R_1, \ldots, S_{T-1}, A_{T-1}, R_T, S_T$
  $G \leftarrow 0$
  $W \leftarrow 1$
  For $t = T-1, T-2, \ldots$ down to 0:
    $G \leftarrow \gamma G + R_{t+1}$
    $C(S_t, A_t) \leftarrow C(S_t, A_t) + W$
    $Q(S_t, A_t) \leftarrow Q(S_t, A_t) + \frac{W}{C(S_t, A_t)} [G - Q(S_t, A_t)]$
    $\pi(S_t) \leftarrow \arg\max_a Q(S_t, a)$ (with ties broken consistently)
    If $A_t \neq \pi(S_t)$ then exit For loop
    $W \leftarrow W \frac{1}{b(A_t|S_t)}$

b

State	P(F)	P(S)
Cool	0.4	0.6
Warm	0.6	0.4
Overheated	-	-



Exits to the next time step without W update!

Monte Carlo

	FAST	SLOW	FAST	SLOW	FAST	SLOW
Q	0	0	-10	0	0	0
π		✔		✔		
C	0	0	1	0	0	0

Ep	**1**
G	**-10**
W	**1**
t	**2**
A_t	F
$\pi(S_t)$	S

Episode 1: C, F, 2, W, S, 1, C, F, 2, W, F, -10, O
G=0

t=3: G_3 = -10
t=2: G_2 = 2 + (0.6)(-10) = -4 #WILL BE used in EVERY Visit

Off-policy MC control, for estimating $\pi \approx \pi_*$

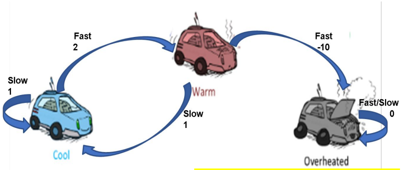
□ **= 0.2**
γ = 0.6

Initialize, for all $s \in \mathcal{S}$, $a \in \mathcal{A}(s)$:
$Q(s,a) \leftarrow$ arbitrary
$C(s,a) \leftarrow 0$
$\pi(s) \leftarrow \text{argmax}_a Q(S_t,a)$ (with ties broken consistently)

Repeat forever:
$b \leftarrow$ any soft policy
Generate an episode using b:
$S_0, A_0, R_1, \ldots, S_{T-1}, A_{T-1}, R_T, S_T$
$G \leftarrow 0$
$W \leftarrow 1$
For $t = T-1, T-2, \ldots$ down to 0:
$G \leftarrow \gamma G + R_{t+1}$
$C(S_t,A_t) \leftarrow C(S_t,A_t) + W$
$Q(S_t,A_t) \leftarrow Q(S_t,A_t) + \frac{W}{C(S_t,A_t)}[G - Q(S_t,A_t)]$
$\pi(S_t) \leftarrow \text{argmax}_a Q(S_t,a)$ (with ties broken consistently)
If $A_t \neq \pi(S_t)$ then exit For loop
$W \leftarrow W \frac{1}{b(A_t|S_t)}$

b

State	P(F)	P(S)
Cool	0.4	0.6
Warm	0.6	0.4
Overheated	-	-



Continue ....

Monte Carlo

Q	**-4**	0	-10	0	0	0
π		✔		✔		
C	**1**	0	1	0	0	0

Ep	**1**
G	**-4**
W	**1**
t	**2**
A_t	F
$\pi(S_t)$	S

Episode 1: C, F, 2, W, S, 1, C, F, 2, W, F, -10, O

G=0

t=3: G_3 = -10

t=2: G_2 = 2 + (0.6)(-10) = -4

Off-policy MC control, for estimating $\pi \approx \pi_*$

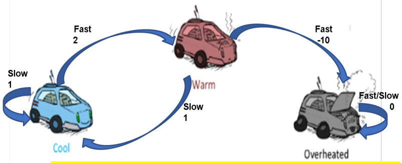
Initialize, for all $s \in \mathcal{S}$, $a \in \mathcal{A}(s)$:
  $Q(s,a) \leftarrow$ arbitrary
  $C(s,a) \leftarrow 0$
  $\pi(s) \leftarrow \mathrm{argmax}_a Q(S_t, a)$ (with ties broken consistently)

□ = 0.2
γ = 0.6

Repeat forever:
  $b \leftarrow$ any soft policy
  Generate an episode using b:
    $S_0, A_0, R_1, \ldots, S_{T-1}, A_{T-1}, R_T, S_T$
  $G \leftarrow 0$
  $W \leftarrow 1$
  For $t = T-1, T-2, \ldots$ down to 0:
    $G \leftarrow \gamma G + R_{t+1}$
    $C(S_t, A_t) \leftarrow C(S_t, A_t) + W$
    $Q(S_t, A_t) \leftarrow Q(S_t, A_t) + \frac{W}{C(S_t, A_t)} [G - Q(S_t, A_t)]$
    $\pi(S_t) \leftarrow \mathrm{argmax}_a Q(S_t, a)$ (with ties broken consistently)
    If $A_t \neq \pi(S_t)$ then exit For loop
    $W \leftarrow W \frac{1}{b(A_t|S_t)}$

b

State	P(F)	P(S)
Cool	0.4	0.6
Warm	0.6	0.4
Overheated	-	-



No change in W! Exit to next t

Monte Carlo



	FAST	SLOW	FAST	SLOW	FAST	SLOW
Q	-4	0	-10	**-1.4**	0	0
π		✔		✔		
C	1	0	1	**1**	0	0

Ep	**1**
G	**-1.4**
W	**1**
t	**1**
A_t	S
$\pi(S_t)$	S

Episode 1: C, F, 2, W, S, 1, C, F, 2, W, F, -10, O

G=0

t=3: G_3 = -10

t=2: G_2 = 2 + (0.6)(-10) = -4

t=1: G_1 = 1+ (0.6*2) + (0.6*0.6*-10) = -1.4

Off-policy MC control, for estimating $\pi \approx \pi_*$

Initialize, for all $s \in \mathcal{S}$, $a \in \mathcal{A}(s)$:
- $Q(s,a) \leftarrow$ arbitrary
- $C(s,a) \leftarrow 0$
- $\pi(s) \leftarrow \text{argmax}_a\, Q(S_t,a)$ (with ties broken consistently)

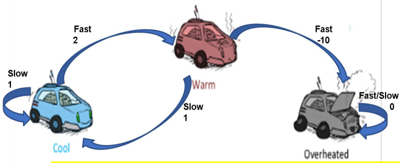
□ = 0.2

γ = 0.6

Repeat forever:
- $b \leftarrow$ any soft policy
- Generate an episode using b:
  - $S_0, A_0, R_1, \ldots, S_{T-1}, A_{T-1}, R_T, S_T$
- $G \leftarrow 0$
- $W \leftarrow 1$
- For $t = T-1, T-2, \ldots$ down to 0:
  - $G \leftarrow \gamma G + R_{t+1}$
  - $C(S_t,A_t) \leftarrow C(S_t,A_t) + W$
  - $Q(S_t,A_t) \leftarrow Q(S_t,A_t) + \frac{W}{C(S_t,A_t)}[G - Q(S_t,A_t)]$
  - $\pi(S_t) \leftarrow \text{argmax}_a\, Q(S_t,a)$ (with ties broken consistently)
  - If $A_t \neq \pi(S_t)$ then exit For loop
  - $W \leftarrow W \frac{1}{b(A_t|S_t)}$

b

State	**P(F)**	**P(S)**
Cool	0.4	0.6
Warm	0.6	0.4
Overheated	-	-



Continue ……..

Monte Carlo

Q	-4	0	-10	**-1.4**	0	0
π		✔		✔		
C	1	0	1	**1**	0	0

Ep	**1**
G	**-1.4**
W	**2.5**
t	**1**
A_t	S
$\pi(S_t)$	S

W = 1/b(S|W)
= 1/0.4
=2.5

Episode 1: C, F, 2, W, S, 1, C, F, 2, W, F, -10, O
G=0

t=3: G_3 = -10
t=2: G_2 = 2 + (0.6)(-10) = -4
t=1: G_1 = 1+ (0.6*2) + (0.6*0.6*-10) = -1.4

Off-policy MC control, for estimating $\pi \approx \pi_*$

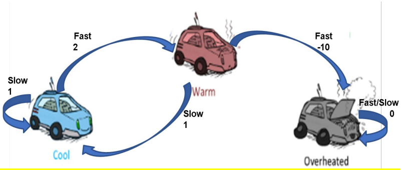
□ = 0.2
γ = 0.6

Initialize, for all $s \in \mathcal{S}$, $a \in \mathcal{A}(s)$:
$Q(s,a) \leftarrow$ arbitrary
$C(s,a) \leftarrow 0$
$\pi(s) \leftarrow \text{argmax}_a Q(S_t, a)$ (with ties broken consistently)

Repeat forever:
$b \leftarrow$ any soft policy
Generate an episode using b:
$S_0, A_0, R_1, \ldots, S_{T-1}, A_{T-1}, R_T, S_T$
$G \leftarrow 0$
$W \leftarrow 1$
For $t = T-1, T-2, \ldots$ down to 0:
$G \leftarrow \gamma G + R_{t+1}$
$C(S_t, A_t) \leftarrow C(S_t, A_t) + W$
$Q(S_t, A_t) \leftarrow Q(S_t, A_t) + \frac{W}{C(S_t, A_t)} [G - Q(S_t, A_t)]$
$\pi(S_t) \leftarrow \text{argmax}_a Q(S_t, a)$ (with ties broken consistently)
If $A_t \neq \pi(S_t)$ then exit For loop
$W \leftarrow W \frac{1}{b(A_t|S_t)}$

b

State	**P(F)**	**P(S)**
Cool	0.4	0.6
Warm	0.6	0.4
Overheated	-	-



At t=1 the best action & current action is same .Hence the Global Weight for WIS is updated

Monte Carlo

	FAST	SLOW	FAST	SLOW	FAST	SLOW
Q	**-3.14**	0	-10	**-1.4**	0	0
π		✔		✔		
C	**3.5**	0	1	**1**	0	0

Ep	1
G	-1.4
W	2.5
t	0
A_t	F
$\pi(S_t)$	S

Off-policy MC control, for estimating $\pi \approx \pi_*$

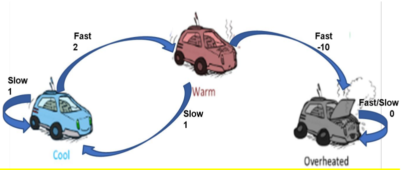
Initialize, for all $s \in \mathcal{S}$, $a \in \mathcal{A}(s)$:
- $Q(s,a) \leftarrow$ arbitrary
- $C(s,a) \leftarrow 0$
- $\pi(s) \leftarrow \operatorname{argmax}_a Q(S_t,a)$ (with ties broken consistently)

□ = 0.2
γ = 0.6

Repeat forever:
- $b \leftarrow$ any soft policy
- Generate an episode using b:
  - $S_0, A_0, R_1, \ldots, S_{T-1}, A_{T-1}, R_T, S_T$
- $G \leftarrow 0$
- $W \leftarrow 1$
- For $t = T-1, T-2, \ldots$ down to 0:
  - $G \leftarrow \gamma G + R_{t+1}$
  - $C(S_t,A_t) \leftarrow C(S_t,A_t) + W$
  - $Q(S_t,A_t) \leftarrow Q(S_t,A_t) + \frac{W}{C(S_t,A_t)}[G - Q(S_t,A_t)]$
  - $\pi(S_t) \leftarrow \operatorname{argmax}_a Q(S_t,a)$ (with ties broken consistently)
  - If $A_t \neq \pi(S_t)$ then exit For loop
  - $W \leftarrow W \frac{1}{b(A_t|S_t)}$

b

State	P(F)	P(S)
Cool	0.4	0.6
Warm	0.6	0.4
Overheated	-	-



Episode 1: $\boxed{\text{C, F, 2, W,}}$ S, 1, C, F, 2, W, F, -10, O

G=0

t=3: G_3 = -10
t=2: G_2 = 2 + (0.6)(-10) = -4
t=1: G_1 = 1+ (0.6*2) + (0.6*0.6*-10) = -1.4
t=0: G_0 = 2+ (0.6*1) + (0.6*0.6*2) + (0.6*0.6*0.6-10) = 1.16

$$Q(C,F) = -4 + \frac{2.5}{3.5}[1.16 - (-4)] = \sim -3.14$$

Continuing to next t=0……..
After update observe that the episode terminates

Monte Carlo

	FAST	SLOW	FAST	SLOW	FAST	SLOW
Q	-3.14	0	-10	-1.4	0	0
π		✔		✔		
C	3.5	0	**2**	1	0	0

Ep	**2**
G	**0 □-10**
W	**1**
t	**2**
A_t	F
$\pi(S_t)$	**S**

Episode 1: C, F, 2, W, S, 1, C, F, 2, W, F, -10, O

Episode 2: C, S, 1, C, F, 2, W, F, -10, O
G=0
t=2: G_2 = -10

$$Q(W,F) = -10 + \frac{1}{2}[-10 - (-10)] = -10$$

Off-policy MC control, for estimating $\pi \approx \pi_*$

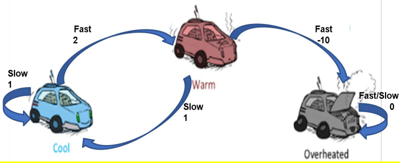
Initialize, for all $s \in \mathcal{S}$, $a \in \mathcal{A}(s)$:
- $Q(s,a) \leftarrow$ arbitrary
- $C(s,a) \leftarrow 0$
- $\pi(s) \leftarrow \operatorname{argmax}_a Q(S_t, a)$ (with ties broken consistently)

□ = 0.2
γ = 0.6

Repeat forever:
- $b \leftarrow$ any soft policy
- Generate an episode using b:
  - $S_0, A_0, R_1, \ldots, S_{T-1}, A_{T-1}, R_T, S_T$
- $G \leftarrow 0$
- $W \leftarrow 1$
- For $t = T-1, T-2, \ldots$ down to 0:
  - $G \leftarrow \gamma G + R_{t+1}$
  - $C(S_t, A_t) \leftarrow C(S_t, A_t) + W$
  - $Q(S_t, A_t) \leftarrow Q(S_t, A_t) + \frac{W}{C(S_t, A_t)}[G - Q(S_t, A_t)]$
  - $\pi(S_t) \leftarrow \operatorname{argmax}_a Q(S_t, a)$ (with ties broken consistently)
  - If $A_t \neq \pi(S_t)$ then exit For loop
  - $W \leftarrow W \frac{1}{b(A_t|S_t)}$

b

State	P(F)	P(S)
Cool	0.4	0.6
Warm	0.6	0.4
Overheated	-	-



Generate new episode from "b" . Continue ……
Results of every time-step in Episode 2 is shown

Monte Carlo

	Cool FAST	Cool SLOW	Warm FAST	Warm SLOW	Overheated FAST	Overheated SLOW
Q	-7.01	0	-10	-1.4	0	0
π		✔		✔		
C	4.5	0	2	1	0	0

Ep	2
G	-10□-4
W	1
t	1
A_t	F
$\pi(S_t)$	S

Episode 1: C, F, 2, W, S, 1, C, F, 2, W, F, -10, O

Episode 2: C, S, 1, $\boxed{\text{C, F, 2, W, F}}$, -10, O

G=0

t=2: G_2 = -10

t=1: G_1 = 2 + (0.6)(-10) = -4

$$Q(W,F) = -3.14 + \frac{4.5}{1}\left[-4 - (-3.14)\right] = -7.01$$

Off-policy MC control, for estimating $\pi \approx \pi_*$

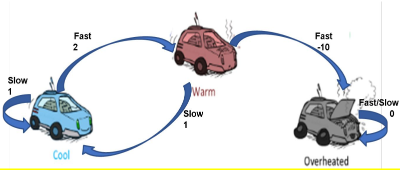
Initialize, for all $s \in \mathcal{S}$, $a \in \mathcal{A}(s)$:
- $Q(s,a) \leftarrow$ arbitrary
- $C(s,a) \leftarrow 0$
- $\pi(s) \leftarrow \operatorname{argmax}_a Q(S_t, a)$ (with ties broken consistently)

□ = 0.2
γ = 0.6

Repeat forever:
- $b \leftarrow$ any soft policy
- Generate an episode using b:
  - $S_0, A_0, R_1, \ldots, S_{T-1}, A_{T-1}, R_T, S_T$
- $G \leftarrow 0$
- $W \leftarrow 1$
- For $t = T-1, T-2, \ldots$ down to 0:
  - $G \leftarrow \gamma G + R_{t+1}$
  - $C(S_t, A_t) \leftarrow C(S_t, A_t) + W$
  - $Q(S_t, A_t) \leftarrow Q(S_t, A_t) + \frac{W}{C(S_t, A_t)}\left[G - Q(S_t, A_t)\right]$
  - $\pi(S_t) \leftarrow \operatorname{argmax}_a Q(S_t, a)$ (with ties broken consistently)
  - If $A_t \neq \pi(S_t)$ then exit For loop
  - $W \leftarrow W \frac{1}{b(A_t|S_t)}$

b

State	P(F)	P(S)
Cool	0.4	0.6
Warm	0.6	0.4
Overheated	-	-



Continue ……
Results of every time-step in Episode 2 is shown

Monte Carlo

	FAST	SLOW	FAST	SLOW	FAST	SLOW
Q	-7.01	**-1.4**	-10	-1.4	0	0
π		✔		✔		
C	4.5	**1**	2	1	0	0

Ep	2
G	-4□-1.4
W	**1.67**
t	0
A_t	S
$\pi(S_t)$	**S**

W = 1/b(S|W)
= 1/0.6
=~1.67

Off-policy MC control, for estimating $\pi \approx \pi_*$

□ = 0.2
γ = 0.6

Initialize, for all $s \in \mathcal{S}$, $a \in \mathcal{A}(s)$:
$Q(s,a) \leftarrow$ arbitrary
$C(s,a) \leftarrow 0$
$\pi(s) \leftarrow \arg\max_a Q(S_t,a)$ (with ties broken consistently)

Repeat forever:
$b \leftarrow$ any soft policy
Generate an episode using b:
$S_0, A_0, R_1, \ldots, S_{T-1}, A_{T-1}, R_T, S_T$
$G \leftarrow 0$
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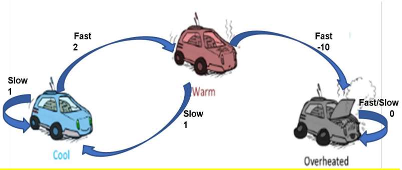
$C(S_t,A_t) \leftarrow C(S_t,A_t) + W$
$Q(S_t,A_t) \leftarrow Q(S_t,A_t) + \frac{W}{C(S_t,A_t)}[G - Q(S_t,A_t)]$
$\pi(S_t) \leftarrow \arg\max_a Q(S_t,a)$ (with ties broken consistently)
If $A_t \neq \pi(S_t)$ then exit For loop
$W \leftarrow W \frac{1}{b(A_t|S_t)}$

b

State	P(F)	P(S)
Cool	0.4	0.6
Warm	0.6	0.4
Overheated	-	-



Episode 1: C, F, 2, W, S, 1, C, F, 2, W, F, -10, O

Episode 2: $\boxed{\text{C, S, 1, C,}}$ F, 2, W, F, -10, O

G=0
t=2: G_2 = -10
t=1: G_1 = 2 + (0.6)(-10) = -4
t=0: G_0 = 1+ (0.6*2) + (0.6*0.6*-10) = -1.4

$$Q(W,F) = 0 + \frac{1}{1}[-1.4 - (0)] = -1.4$$

This completes processing second episode. Continue next episode

Monte Carlo Methods

1. Given
   - S, A, optionally R(s, a, s`)
2. To find
   - Policy π . This is referred to as **TARGET** policy
3. Solution
   - Sample Episodes
   - Compute estimates of the value by discounted returns
     - ✔ Simple average
     - ✔ Weighted average (OIS : Ordinary Importance Sampling)
     - ✔ Normalized weighted average (WIS : Weighted Importance Sampling)
4. Idea
   - On Policy :
     - Generate Sample/Experience Evaluate using Return Improve
   - Off Policy :
     - Refer to Sample/Experience generated by another reference(**BEHAVIOUR**) policy Evaluate using Return Improve

Monte Carlo



	FAST	SLOW	FAST	SLOW	FAST	SLOW
Q	-7.01	-1.4	-10	-1.4	0	0
π		✔		✔		
C	4.5	1	2	1	0	0

Ep	
G	
W	
t	
A_t	
$\pi(S_t)$	

Episode 1: C, F, 2, W, S, 1, C, F, 2, W, F, -10, O

Episode 2: C, S, 1, C, F, 2, W, F, -10, O

Important Note : The Ordinary Importance Sampling (OIS) is unbiased but will produce high variance. The same given algorithm can be adapted to perform ordinary importance sampling by changing only the count update equation to : C(St,At) □ C(St,At) + 1

Off-policy MC control, for estimating $\pi \approx \pi_*$

□ **= 0.2**
γ = 0.6

Initialize, for all $s \in \mathcal{S}$, $a \in \mathcal{A}(s)$:
  $Q(s,a) \leftarrow$ arbitrary
  $C(s,a) \leftarrow 0$
  $\pi(s) \leftarrow \operatorname{argmax}_a Q(S_t,a)$ (with ties broken consistently)

Repeat forever:
  $b \leftarrow$ any soft policy
  Generate an episode using b:
    $S_0, A_0, R_1, \ldots, S_{T-1}, A_{T-1}, R_T, S_T$
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  For $t = T-1, T-2, \ldots$ down to 0:
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    $Q(S_t,A_t) \leftarrow Q(S_t,A_t) + \frac{W}{C(S_t,A_t)}[G - Q(S_t,A_t)]$
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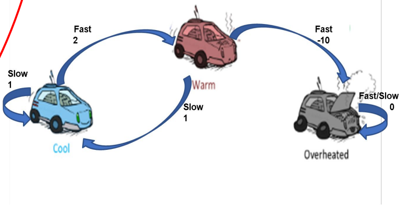
    If $A_t \neq \pi(S_t)$ then exit For loop
    $W \leftarrow W \frac{1}{b(A_t|S_t)}$

b

State	P(F)	P(S)
Cool	0.4	0.6
Warm	0.6	0.4
Overheated	-	-



Monte Carlo

Q	-7.01	-1.4	-10	-1.4	0	0
π		✔		✔		
C	4.5	1	2	1	0	0

Ep	**3**
G	**0**
W	**1**
t	
A_t	
$\pi(S_t)$	

Off-policy MC control, for estimating $\pi \approx \pi_*$

□ = 0.2
γ = 0.6

Initialize, for all $s \in \mathcal{S}$, $a \in \mathcal{A}(s)$:
 $Q(s,a) \leftarrow$ arbitrary
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Repeat forever:
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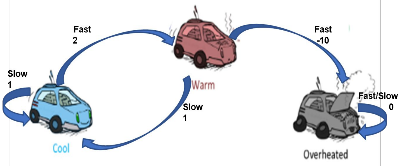
b

State	P(F)	P(S)
Cool	0.4	0.6
Warm	0.6	0.4
Overheated	-	-

Episode 1: C, F, 2, W, S, 1, C, F, 2, W, F, -10, O

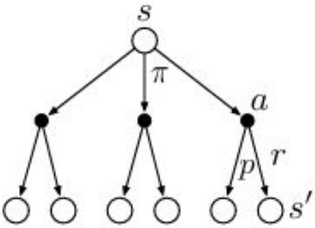
Episode 2: C, S, 1, C, F, 2, W, F, -10, O

Episode 3: C, S, 1, C, S, 1, C, F, 2, W, F, -10, O

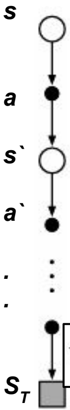


Students are advised to take this episode #3 as an exercise for practice …………

DP VS MC

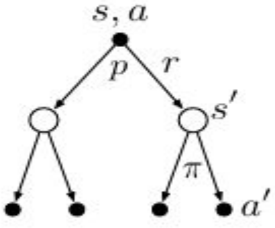


$$V(s) \leftarrow \sum_a \pi(a|s) \sum_{s',r} p(s',r|s,a)\left[r + \gamma V(s')\right]$$

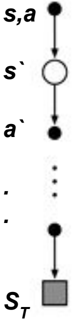


$$V_{n+1} \doteq V_n + \frac{W_n}{C_n}\left[G_n - V_n\right]$$

DP VS MC



$$q_\pi(s,a) = \sum_{s',r} p(s',r|s,a)\left[r + \gamma \sum_{a'} \pi(a'|s')q_\pi(s',a')\right]$$



$$Q(S_t,A_t) \leftarrow Q(S_t,A_t) + \frac{W}{C(S_t,A_t)}\left[G - Q(S_t,A_t)\right]$$